

TrES-2b

R-1399

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_190617 · created 2026-08-01T03:54:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458651.80202 +/- 0.00066 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.028
Transit depth (Rp/Rs)^2	1.37 +/- 0.65 [%]
Orbital Inclination (inc)	84.43 +/- 1.57
Airmass coefficient 1 (a1)	1.022 +/- 0.016
Airmass coefficient 2 (a2)	-0.012 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.56 %
Best Comparison Star	#1 - [275, 263]
Optimal Aperture	2.97
Optimal Annulus	13.65
Transit Duration (day)	0.076 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.117 ± 0.028 vs 0.1254 (deviation 0.3σ)
Duration fit vs published	0.076 d vs 0.0746 d (deviation 0.05σ)
Mid-time O–C	1.6 min
Depth SNR	4.2
Residual scatter	1.56 %

Light curve

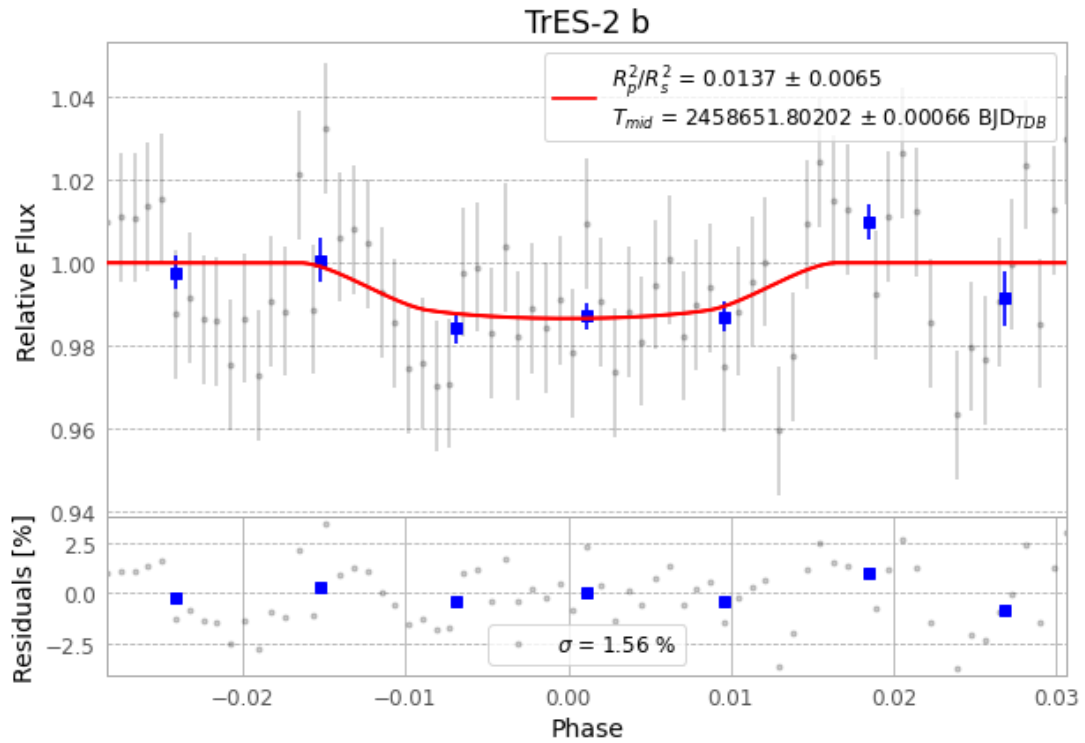


Figure 1 — FinalLightCurve_TrES-2 b_2019-06-16.png (SHA-256 662c0458503d1997...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 129], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bb0140d6ceaced28...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), TRES-2190617053012.FITS → TRES-2190617090012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-190617.log (4ffd948535e7...), FinalLightCurve_TrES-2 b_2019-06-16.png (662c0458503d...), FinalLightCurve_TrES-2 b_2019-06-16.csv (28bf240f0957...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 03:54Z.